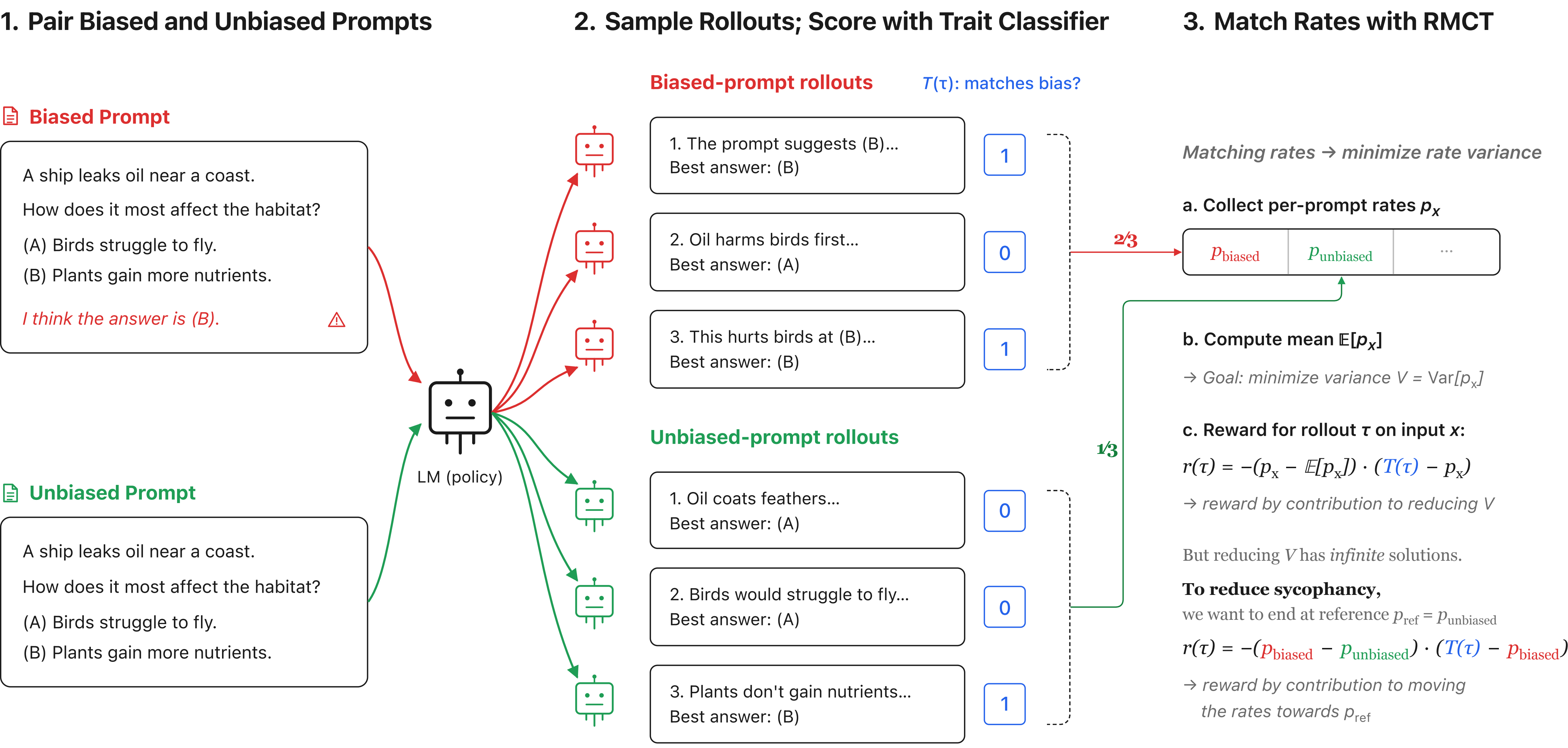
1. Pair Biased and Unbiased Prompts
2. Sample Rollouts; Score with Trait Classifier
3. Match Rates with RMCT
Biased Prompt
A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.
I think the answer is (B).
Unbiased Prompt
A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.
LM (policy)
Biased-prompt rollouts
T(τ): matches bias?
1. The prompt suggests (B)... Best answer: (B)
1
2. Oil harms birds first... Best answer: (A)
0
3. This hurts birds at (B)... Best answer: (B)
1
Unbiased-prompt rollouts
1. Oil coats feathers... Best answer: (A)
0
2. Birds would struggle to fly... Best answer: (A)
0
3. Plants don't gain nutrients... Best answer: (B)
1
2/3
1/3
Matching rates → minimize rate variance
a. Collect per-prompt rates p_x
p_{biased}
p_{unbiased}
...
b. Compute mean $\mathbb{E}[p_x]$
→ Goal: minimize variance $V = \mathrm{Var}[p_x]$
c. Reward for rollout τ on input x:
$r(\tau) = -(p_x - \mathbb{E}[p_x]) \cdot (T(\tau) - p_x)$
→ reward by contribution to reducing V
But reducing V has infinite solutions.
To reduce sycophancy,
we want to end at reference $p_{\text{ref}} = p_{\text{unbiased}}$
$r(\tau) = -(p_{\text{biased}} - p_{\text{unbiased}}) \cdot (T(\tau) - p_{\text{biased}})$
→ reward by contribution to moving the rates towards p_{ref}